Operational Analysis of Flight Delays and Diversions

A Data-Driven Investigation of Operational Drivers

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List of Acronyms

- ArrDelay: Arrival Delay (minutes)
- CRS: Computer Reservation System
- CRSArrTime: Scheduled Arrival Time
- CRSDepTime: Scheduled Departure Time
- DepDelay: Departure Delay (minutes)
- DepHour: Scheduled Departure Hour
- GLM: Generalised Linear Model
- OR: Odds Ratio
- PlaneAge: Aircraft Age (years)

I- Introduction

Air transportation systems generate large volumes of operational data that can be used to understand patterns of delays and disruptions. Identifying the factors that influence flight delays and diversions is particularly important for improving scheduling efficiency, airport operations, and passenger experience. In modern airline systems, even small delays can propagate through flight networks, affecting subsequent departures and arrivals. Consequently, statistical analysis of operational data can provide valuable insights into the mechanisms that generate delays and operational disruptions.

The Analysis investigates flight performance using a subset of five consecutive years (2003–2007) from the ASA airline dataset. The objective is to uncover patterns in delays and better understand the factors influencing flight reliability.

In particular, the analysis focuses on identifying the best times and days of the week to minimize delays, evaluating whether aircraft age has an impact on delays, and modelling the probability of flight diversions using logistic regression.

II- Data Cleaning and pre-processing

Prior to analysis, several data cleaning and preprocessing steps were applied to ensure consistency and reliability of the results. Given the size of the dataset, each year was processed separately before combining the results where appropriate. Only the variables relevant to each part of the analysis were retained, which helped reduce memory usage and improve computational efficiency. Flights that were cancelled were removed from the dataset, as they do not provide meaningful information about delays. Observations with missing values in key variables such as arrival delay, departure delay, and scheduled departure time were also excluded to maintain the integrity of the analysis. In addition, the scheduled departure time was transformed into an hourly format by dividing the scheduled departure time (CRSDepTime) variable by 100.

This transformation enabled a more interpretable analysis of delay patterns throughout the day. For the aircraft age analysis, flight records were merged with an external dataset containing aircraft manufacturing years using the tail number as a common identifier. Aircraft age was then computed as the difference between the year of the flight and the year of manufacture.

III- Analysis and Results

1- Best Times and Days to Minimize Delays

To identify the most reliable travel times, average arrival delays were computed by hour of departure and day of the week for each year.

The results are presented in two complementary visuals. **Figure 1** reports average arrival delay by scheduled departure hour, while **Figure 2** reports average arrival delay by day of the week. Scheduled departure time is used because it provides a consistent planned timing variable available before the flight takes place, making it suitable for comparing delay patterns across flights and years.

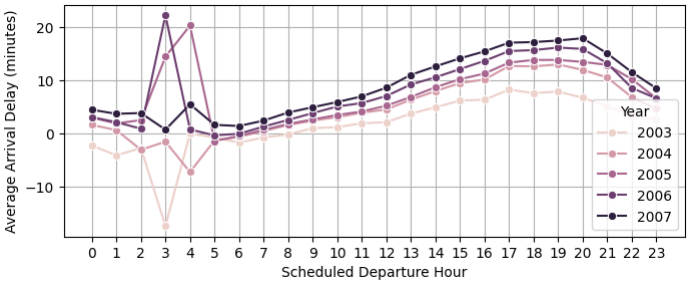


Figure 1: Average Arrival Delay by Scheduled Departure Hour (2003–2007)

As shown in **Figure 1**, a clear and consistent pattern emerges across all five years. Flights scheduled during the early hours of the day, particularly between midnight and approximately 7 a.m., experience the lowest delays. In several instances, the average delay is negative, which indicates that flights in those time periods tend, on average, to arrive slightly earlier than scheduled. However, this should be interpreted cautiously, since delays are also influenced by other factors such as route characteristics, operational conditions, and congestion effects accumulating throughout the day. As the day progresses, delays gradually increase, reaching their peak in the late afternoon and evening, typically between 5 p.m. and 9 p.m.

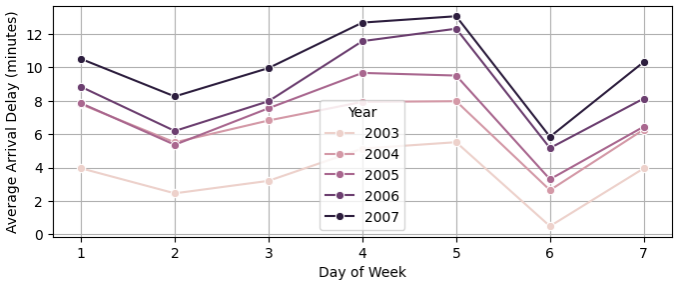


Figure 2: Average Arrival Delay by Day of the Week (2003–2007)

A similar pattern is observed when examining delays by day of the week. **Figure 2** shows that while the optimal day varies slightly across years, certain days consistently maintain lower average delays. However, compared to the time-of-day effect, the variation across days is less pronounced and less stable across years.

Additional visualizations, including year-specific plots and supporting heatmaps, are provided in **Appendix A.1** and **Appendix A.2**, which further confirm the consistency of these patterns across all years.

Overall, these results indicate that the time of departure plays a dominant role in determining flight delays. Early morning flights are significantly more reliable, likely due to lower congestion and the absence of accumulated delays from earlier flights.

2- Impact of Aircraft Age on Delays

To investigate whether aircraft age influences delays, flight data were merged with aircraft manufacturing information, allowing the age of each aircraft to be computed. The relationship between aircraft age and average arrival delay is illustrated in **Figure 3**, which plots mean arrival delay across aircraft ages over the full period.

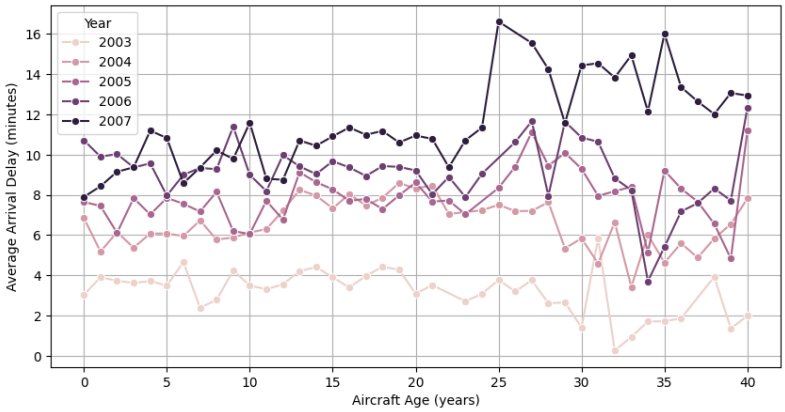


Figure 3: Average Arrival Delay as a Function of Aircraft Age (2003–2007)

As shown in **Figure 3**, there is a weak positive relationship between aircraft age and arrival delay. Older aircraft tend to exhibit slightly higher delays on average; however, the relationship is not strong and does not remain consistent across all age groups. In particular, variability increases for higher aircraft ages, which is likely due to the smaller number of observations associated with very old aircraft, making estimates less stable.

Additional supporting visualisations are provided in **Appendix A.3** and **Appendix A.4**, which further illustrate the distribution of delays and the variability across aircraft ages.

These findings suggest that aircraft age alone is not a major determinant of delays. While older aircraft may be associated with slightly higher delays, operational factors such as scheduling efficiency, airline practices, and airport congestion are likely to have a more significant impact.

3- Logistic Regression Analysis of Flight Diversions

To analyse the factors influencing flight diversions, a logistic regression model was fitted separately for each year. The dependent variable indicates whether a flight was diverted, and the explanatory variables include temporal, operational, and airline-specific characteristics such as month, day of the week, scheduled departure and arrival times, distance, departure delay, and carrier.

These variables were selected based on their potential relevance to flight operations. Temporal variables capture seasonal and weekly effects, while operational variables such as departure delay and distance reflect flight conditions. The inclusion of carrier allows for airline-specific differences in diversion behaviour to be captured.

The estimated coefficients for one representative year are presented in **Figure 4**, while additional year-specific results are provided in **Appendix A.5**.

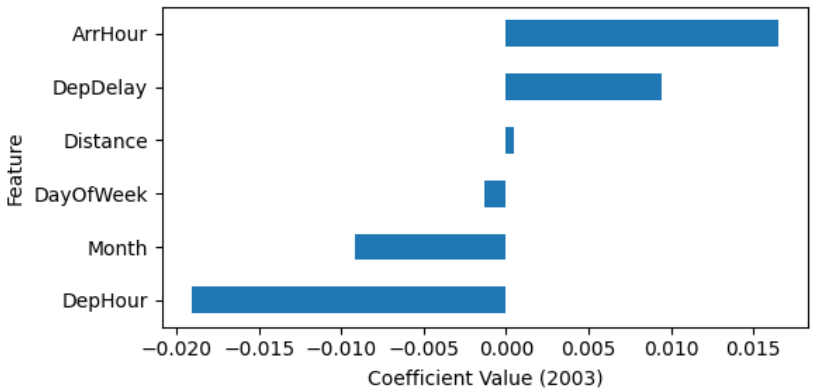


Figure 4: Logistic Regression Coefficients for Predicting Flight Diversions (2003)

The sign of each coefficient indicates the direction of the relationship with the probability of diversion. In particular, departure delay (DepDelay) consistently exhibits a positive coefficient across all years, indicating that flights departing later than scheduled are more likely to be diverted. This suggests that operational disruptions prior to departure significantly increase the risk of diversion.

To better interpret the magnitude of these effects, odds ratios were computed from the estimated coefficients. The results confirm that increases in departure delay substantially raise the likelihood of diversion, highlighting its importance as a key predictor. In comparison, the effects of other variables are relatively small.

Overall, the results are broadly consistent across years, suggesting that the relationship between operational delays and diversion risk is stable over time.

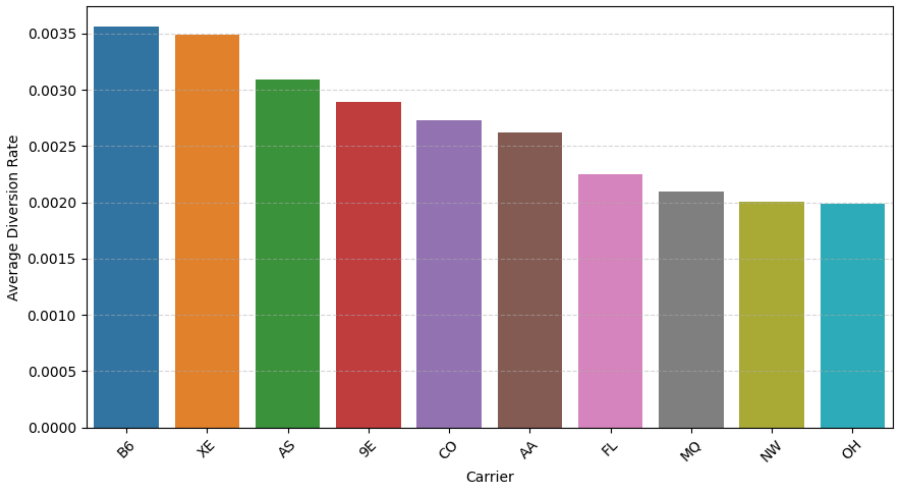


Figure 5: Airlines with the Highest Average Flight Diversion Rates (2003–2007)

In addition to the regression analysis, carrier-level diversion rates were examined to provide further insight into airline-specific behaviour. The results are presented in **Figure 5**, which shows the airlines with the highest average diversion rates over the study period. Although diversion events are relatively rare, there is noticeable variation across carriers, suggesting that airline-specific operational factors may influence diversion likelihood.

IV- Conclusion and perspectives

This analysis highlights several key insights into flight delays and diversions. First, the time of day is a critical factor in determining flight reliability, with early morning departures consistently experiencing the lowest delays. In contrast, delays accumulate throughout the day, leading to significantly higher delays in the evening.

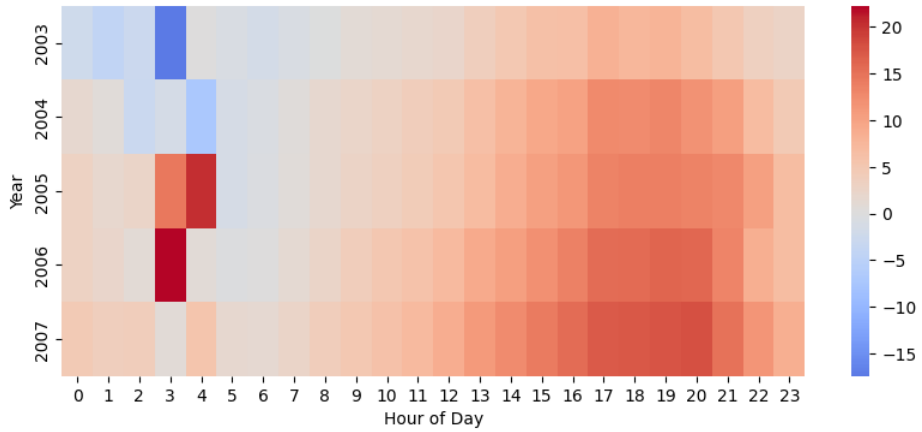
Second, while there is a slight tendency for older aircraft to experience higher delays, the relationship is weak and inconsistent. This suggests that aircraft age is not a primary driver of delays, and that operational and systemic factors are more influential.

Finally, the logistic regression analysis demonstrates that predicting flight diversions is challenging due to the rarity of such events. Although the model identifies some relevant factors, its predictive performance is limited by class imbalance, indicating that more advanced modelling approaches may be required.

Overall, the findings suggest that improving flight reliability is more closely related to managing operational efficiency and system congestion than to factors such as aircraft age alone.

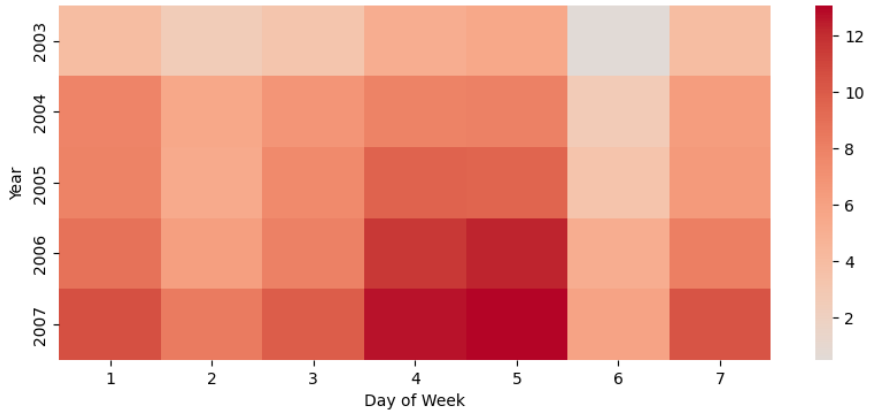
V- Appendix

A.1:



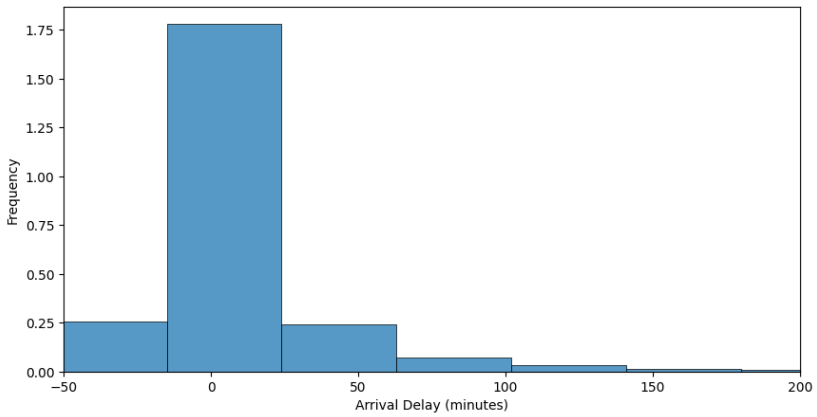
Average Arrival Delay by Scheduled Departure Hour for Each Year (2003–2007)

A.2:



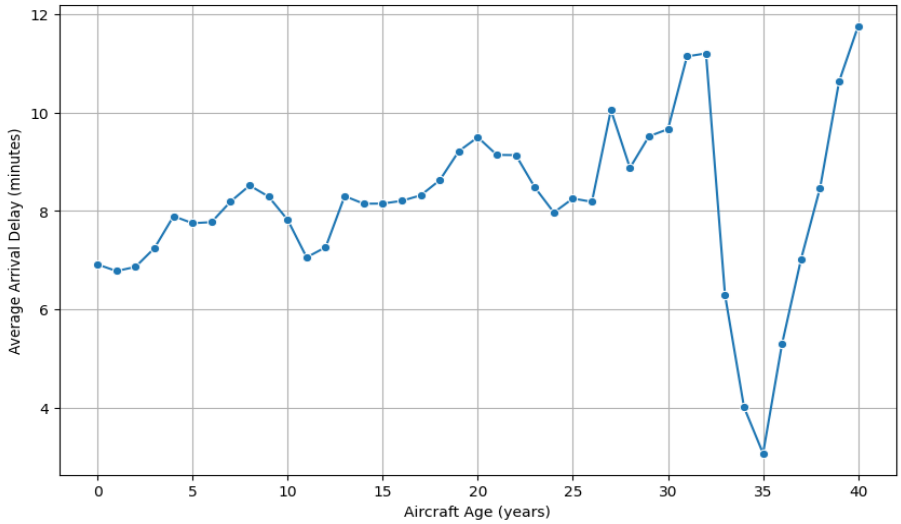
Average Arrival Delay by Day of the Week for Each Year (2003–2007)

A.3:



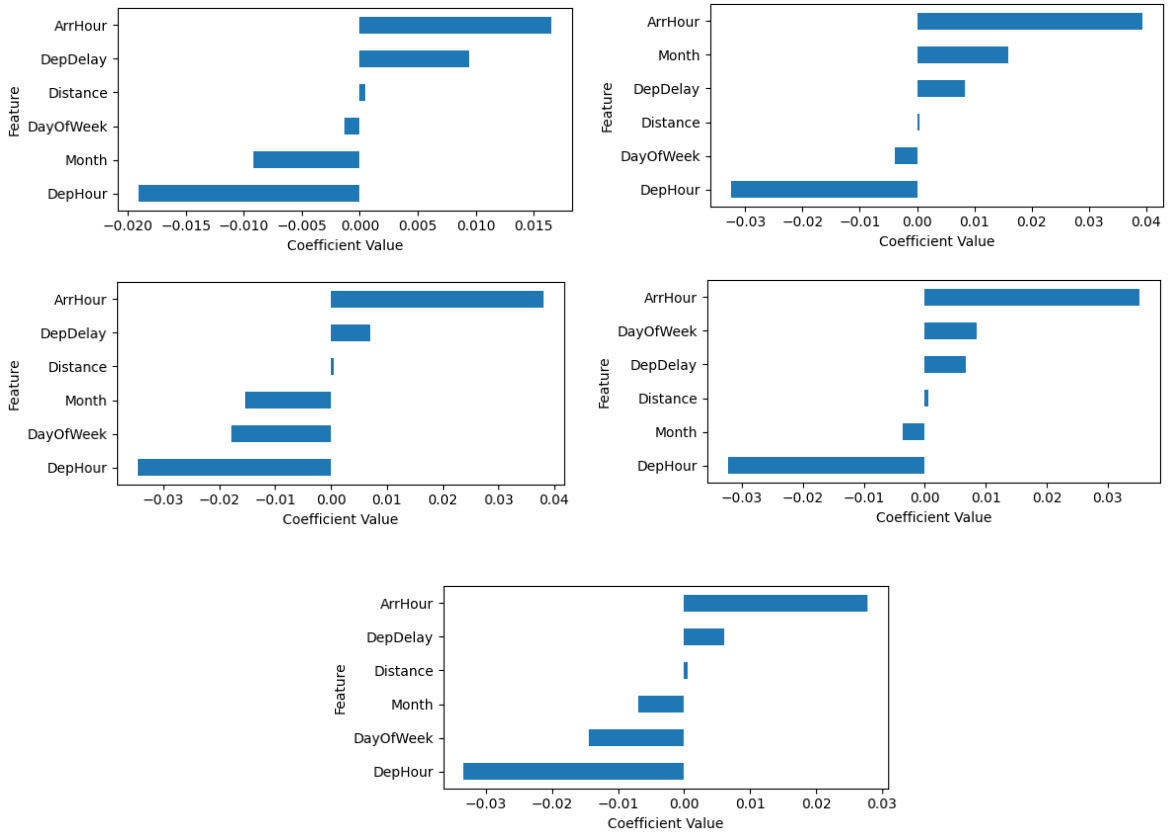
Distribution of Arrival Delays (2003–2007)

A.4:



Average Arrival Delay by Aircraft Age (2003–2007)

A.5:



Logistic Regression Coefficients by Year (2003–2007)